MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER
TYPE 2N5241, JAN, JANTX, AND JANTXV.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, high-power transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-3).

1.3 Maximum ratings.

P_T 1/ T_C = +25°C	P_T 1/ T_C = +100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_{stg} and T_{op}
W	W	V dc	V dc	V dc	A dc	A dc	°C
150	85.7	400	400	6	10	5	-65 to +200

1/ Derate linearly 0.857 W/°C for T_C > 25°C.

1.4 Primary electrical characteristics at T_C = 25°C ±3°C.

Limit	h_{FE} 1/ V_{CE} = 5.0 V dc I_C = 2.5 A dc	$V_{CE(sat)}$ 1/ I_C = 2.5 A dc I_B = 0.5 A dc	C_{obo} V_{CB} = 10 V dc, I_E = 0 100 KHz≤f≤1 MHz	$\lvert h_{fe} \rvert$ I_C = 0.2 A dc V_{CE} = 10 V dc f = 1 MHz	θ_{J-C}
		V dc	pF		°C/W
Min	15	---	---	2.5	---
Max	35	0.7	500	12.5	1.17

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RBE-2), Griffiss AFB, New York, 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold, tin, or solder. Where a choice of lead material or finish is desired, it shall be specified in the contract or order.

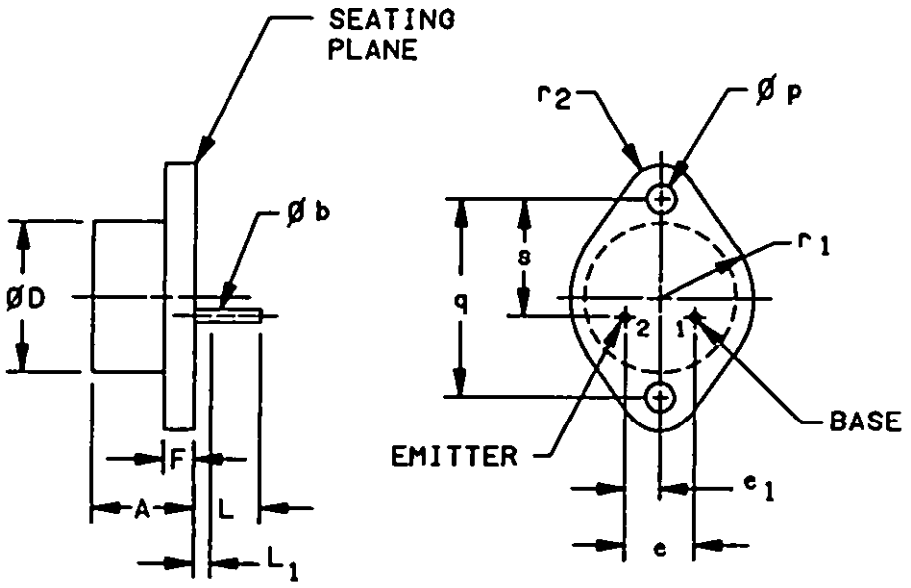
3.4 Marking. Devices shall be marked in accordance with MIL-S-19500. At the option of the manufacturer, the country of origin may be obmitted from the body of the device.

QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

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SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	.250	.328	6.35	8.33	
Øb	.038	.043	.97	1.09	2,5
ØD		.875		22.23	
e	.420	.440	10.67	11.18	
e_1	.205	.225	5.21	5.72	
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	2
L_1		.050		1.27	2,4
Øp	.151	.161	3.84	4.09	
q	1.177	1.197	29.90	30.40	
r_1	.495	.525	12.57	13.34	
r_2	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	1

NOTES:
1. These dimensions should be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gage is not used, measurement will be made at seating plane.
2. Two leads.
3. Collector shall be electrically connected to the case.
4. Diameter of leads in the zone is not controlled.
5. Øb applies between L_1 and L.
6. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Physical dimensions.

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Screen (see table II of MIL-S-19500)	Measurements
	JANTXV and JANTX levels
4	See 4.5.5.
9	I_{CEX1}
11	I_{CEX1} and h_{FE2}; ΔI_{CEX1} = 100% of initial value or .05 mA dc, whichever is greater.
12	See 4.2.1
13	Subgroup 2 of table I herein; ΔI_{CEX1} = 100% of initial value or .05 mA dc, whichever is greater; Δh_{FE2} = +25%, -15% of initial value

4.2.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5°C ±12.5°C V_{CB} = 35±5 V dc $T_A \leq 100°C$

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and table II herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 2.50 A dc.
b. Collector to emitter voltage magnitude shall be 20 V dc.
c. Reference temperature measuring point shall be the case.
d. Reference point temperature shall be 25°C $\leq T_R \leq$ 75°C and shall be recorded before the test is started.
e. Mounting arrangement shall be with heat sink to header.
f. Maximum limit for $R_{\theta JC}$ shall be 1.17°C/W.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions	JAN TX TXV		Min	Max	
Subgroup 1			5				
Visual and mechanical examination	2071						
Subgroup 2			5				
Collector to emitter sustaining voltage	---	I_C = 100 mA; L = 50 mH; f = 30 - 60 Hz (see figure 2)		$V_{(BR)CEO}$	325		V dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6.0 V dc		I_{EBO}		2.0	mA dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 400 V dc		I_{CEO}		0.5	mA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -1.5 V dc; V_{CE} = 400 V dc		I_{CEX1}		0.5	mA dc
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 A dc; pulsed (see 4.5.1)		h_{FE1}	25	---	---
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 2.5 A dc; pulsed (see 4.5.1)		h_{FE2}	15	35	---
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 3.5 A dc; pulsed (see 4.5.1)		h_{FE3}	10	---	---
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5 A dc; pulsed (see 4.5.1)		h_{FE4}	5	---	---
Base to emitter voltage (saturated)	3066	Test condition A; I_B = 0.5 A dc; I_C = 2.5 A dc; pulsed (see 4.5.1)		$V_{BE(sat)1}$	---	1.5	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_B = 1 A dc; I_C = 5 A dc; pulsed (see 4.5.1)		$V_{BE(sat)2}$	---	2.0	V dc
Collector to emitter voltage (saturated)	3071	I_C = 2.5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)		$V_{CE(sat)1}$	---	0.7	V dc
Collector to emitter voltage (saturated)	3071	I_C = 5 A dc; I_B = 1 A dc; pulsed (see 4.5.1)		$V_{CE(sat)2}$	---	2.0	V dc

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions	JAN TX TXV		Min	Max	
Subgroup 3			7				
High temperature operation:		T_A = 150°C					
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -1.5 V dc; V_{CE} = 400 V dc		I_{CEX2}	---	2.5	mA dc
Low temperature operation:		T_A = -55°C					
Forward current tramsfer ratio	3076	V_{CE} = 5 V dc; I_C = 2.5 A dc; pulsed (see 4.5.1)		h_{FE5}	10	---	---
Subgroup 4			7				
Magnitude of common emitter small-signal short-circuit forward-current gain ratio	3306	V_{CE} = 10 V dc; I_C = 0.2 A dc; f = 1 MHz		$\lvert h_{fe} \rvert$	2.5	12.5	---
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 KHz $\leq$ f $\leq$ 1 MHz		C_{obo}	---	500	pF
Pulse response:							
Turn-on time	3251	Test cond. A; V_{CC} = 125 V dc; I_C = 2.5 A dc; I_{B1} = 0.25 A dc; I_{B2} = -0.5 A dc; (see figure 3)		t_{on}	---	0.8	µs
Turn-off time	3251	Test cond. A; V_{CC} = 125 V dc; I_C = 2.5 A dc; I_{B1} = 0.25 A dc; I_{B2} = -0.5 A dc; (see figure 3)		t_{off}	---	1.7	µs
Subgroup 5			10				
Safe operating area (continuous dc)	3051	T_C = 25 +10,-0°C; t $\geq$ 1s; 1 cycle (see figure 4)					
Test 1		I_C = 10 A dc; V_{CE} = 15 V dc;					
Test 2		I_C = 2.14 A dc; V_{CE} = 70 V dc;					
Test 3		I_C = 190 mA dc; V_{CE} = 200 V dc;					

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD JAN TX TXV	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 5 - Continued			10				
Safe operating area (clamped inductive)	---	T_C = 25°C; I_C = 10 A dc; V_{CC} = 15 V dc (see figures 6 and 7); R_L < 2.5 Ω; V_{CC} = 25 V dc; Clamp voltage = 400 +0, -5 V dc; device fails if clamp voltage is not reached.					
End point electrical measurements		Duty cycle < 10% See table IV, steps 1 and 4					
Subgroup 6 and 7							
Not applicable							

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TABLE II. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			15
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			10
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
a. Fine leak b. Gross leak			
Electrical measurements		See table IV, steps 1, and 3.	
Subgroup 3			5
Steady-state operation life	1027	T_J = 187.5°C ± 12.5°C; V_{CB} = 35 ± 5 V dc; $T_A \leq$ 100°C	
Electrical measurements		See table IV, steps 1, 3, 4, 5,and 6.	
Subgroup 4			
Decap internal visual design verification	2075		1 device/0 failure for each lot
Bond strength	2037	Test conditon A; all internal wires for each device shall be pulled separately.	20 (c = 1)
Subgroup 5			
Thermal resistance	3131	See 4.5.2	15
Subgroup 6			7
High temperature life (non-operating)	1032	T_A = 200°C	
Electrical measurements		See table IV, steps 1, 4, 5,and 6.	
Subgroup 7			10
Safe operating area (clamped inductive) (destructive)		T_C = 25°C; I_C = 10 A dc; V_{CC} = 375 V dc (see figures 6 and 7); $R_L \leq$ 2.5 Ω; Clamp voltage = 400 $^{+0}_{-5}$ V dc; device fails if clamp voltage is not reached. Duty cycle ≤ 10%	

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TABLE II. Group B inspection for JAN, JANTX, and JANTXV devices - Continued.

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 7 - Continued			10
Safe operating area (switching)	3053	Load condition C; (unclamped inductive load) (see figure 5); T_C = 25°C; duty cycle ≤ 10% R_S = 0.1 Ω	
Test 1		Tp≈ 5 ms (vary to obtain I_C); R_{BB1} = 4Ω; V_{BB1} = 10 V dc; R_{BB2} = ∞; V_{BB2} = 0; V_{CC} = 25 V dc; I_C = 10 A dc; L = 10 mH; Q ≥ 150 @ 1 KHz, R≈ 0.25 Ω; f_r ≈ 100 KHz	
Test 2		Tp≈ 5 ms (vary to obtain I_C); R_{BB1} = 20Ω; V_{BB1} = 10 V dc; R_{BB2} = ∞; V_{BB2} = 0; V_{CC} = 25 V dc; I_C = 3.4 A dc; L = 100 mH, Q ≥ 150 @ 1 KHz, R≈ 1.0 Ω; f_r ≈ 30 KHz	
End point electrical measurements		See table IV, steps 1 and 4.	
Subgroup 8			5
Constant acceleration	2006	One minute minimum in the Y1 orientation at 30,000G minimum. Only required when lot complies with 4.5.5.	
Electrical measurements		See table IV, steps 1, 2, and 4	

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TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 1			15
Physical dimensions	2066	(See figure 1)	
Subgroup 2			10
Thermal shock (glass strain)	1056	Test condition B	
Terminal strength (tension)	2036	Test condition A; weight = 10 lbs; t = 15s.	
Hermetic seal	1071		
a. Fine leak b. Gross leak			
Moisture resistance	1021	Omit initial conditioning	
External visual	2071		
Electrical measurements		See table IV, steps 1, 2, and 4	
Subgroup 3			10
Shock	2016		
Vibration, variable frequency	2056		
Constant acceleration	2006		
Electrical measurements		See table IV, steps 1, 2, and 4 (JAN, JANTX and JANTXV).	
Subgroup 4			15
Salt atmosphere (corrosion)	1041		
Subgroup 5			15
Barometric pressure	1001	Normal mounting; pressure = 8 ±2 mm Hg for 60s (min)	
Subgroup 6			10
Steady state operation life	1027	V_{CE} = 30 V dc; P_T = 85.7 W at T_C = 100°C or 100°C ≤ T_C ≤ 125°C with P_T varied, according to the chosen T_C, to achieve T_J = 187.5°C ±12.5°C. (see 4.5.3).	
Electrical measurements		See table IV, steps 3, 5, 6, and 7	

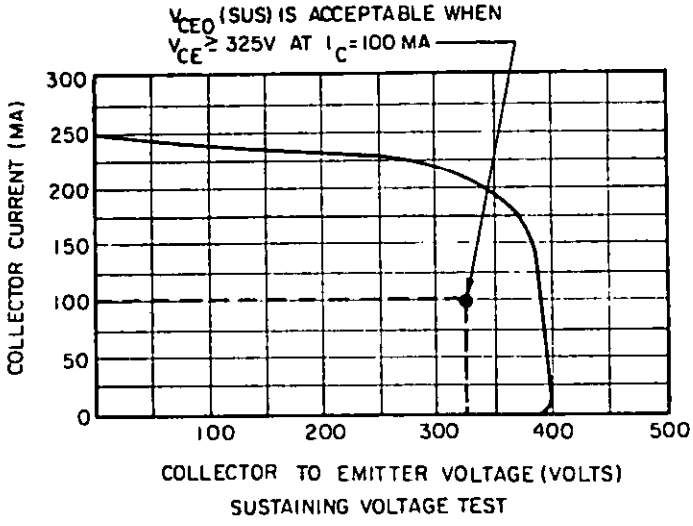
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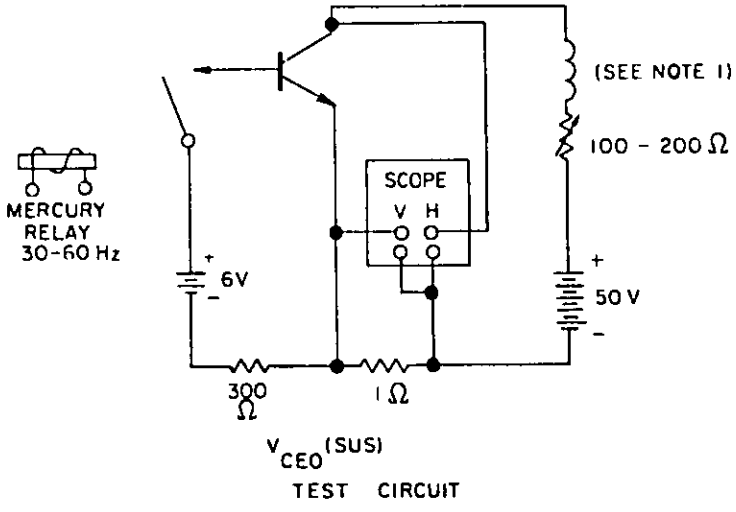
TABLE IV. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -1.5 V dc, V_{CE} = 400 V dc	I_{CEX1}		0.5	mA dc
2.	Collector to emitter voltage (saturated)	3071	I_C = 2.5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$	---	0.7	V dc
3.	Base to emitter voltage (saturated)	3066	Test condition A; I_C = 2.5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$	---	1.5	V dc
4.	Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 2.5 A dc; pulsed (see 4.5.1)	h_{FE2}	15	35	---
5.	Repeat step 1	3041	Same as step 1	ΔI_{CEX1} 1/		100% of initial value or 100 µA dc, whichever is greater.	
6.	Repeat step 4	3076	Same as step 4	Δh_{FE2} 1/	---	±20% change from previously measured value.	
7.	Repeat step 2	3071	Same as step 2	$\Delta V_{CE(SAT)1}$ 1/		±50 mV dc change from previously measured value.	

1/ Devices that exceed the group A limits for this test shall not be acceptable.

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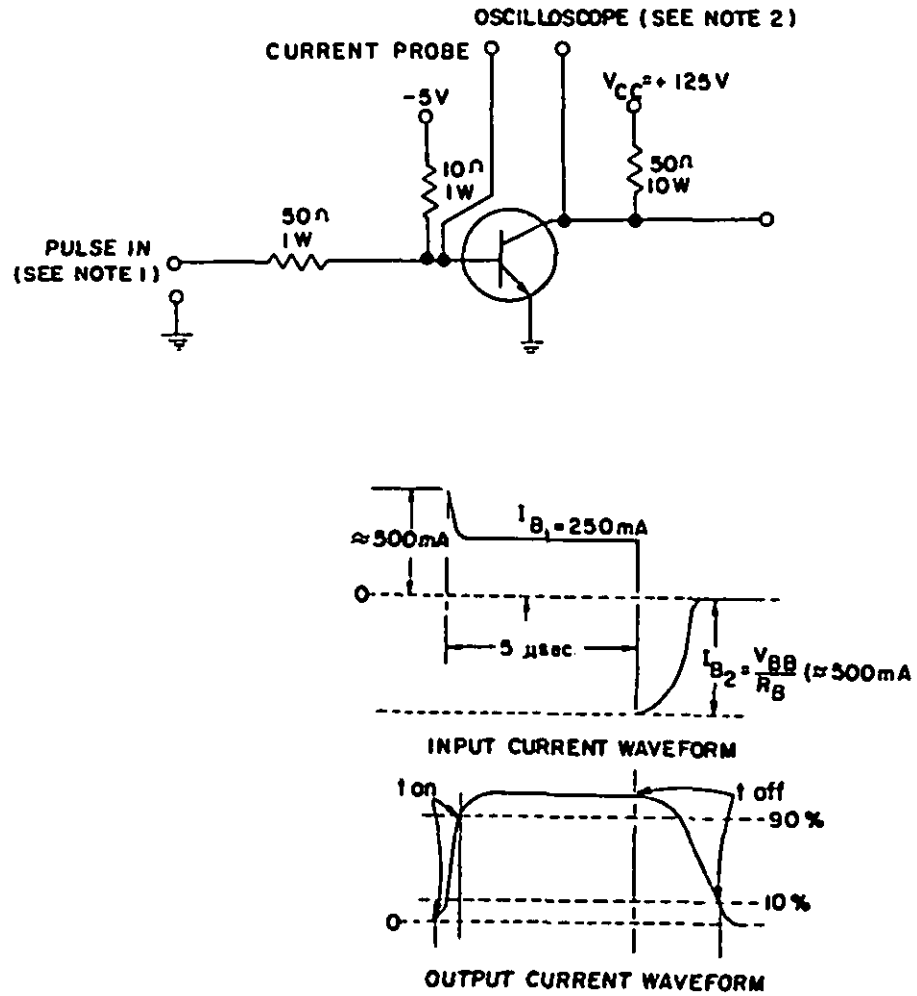




NOTE:
1. L = 50 mH, 2.3Ω, Q = 150 min. resonant frequency 10 kHz (general radio type 940F or equivalent).

FIGURE 2. Sustaining voltage test.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 20 nanoseconds: duty cycle ≤ 5%; generator source impedance shall be 50Ω; pulse width = 5 µs.
2. Oscilloscope:
   Z_{in} ≥ 100 kΩ; C_{in} ≤ 50pF; rise time ≤ 20 ns.
   Current Probe: Tektronix P 6016 or equivalent.

FIGURE 3. Pulse response test circuit.

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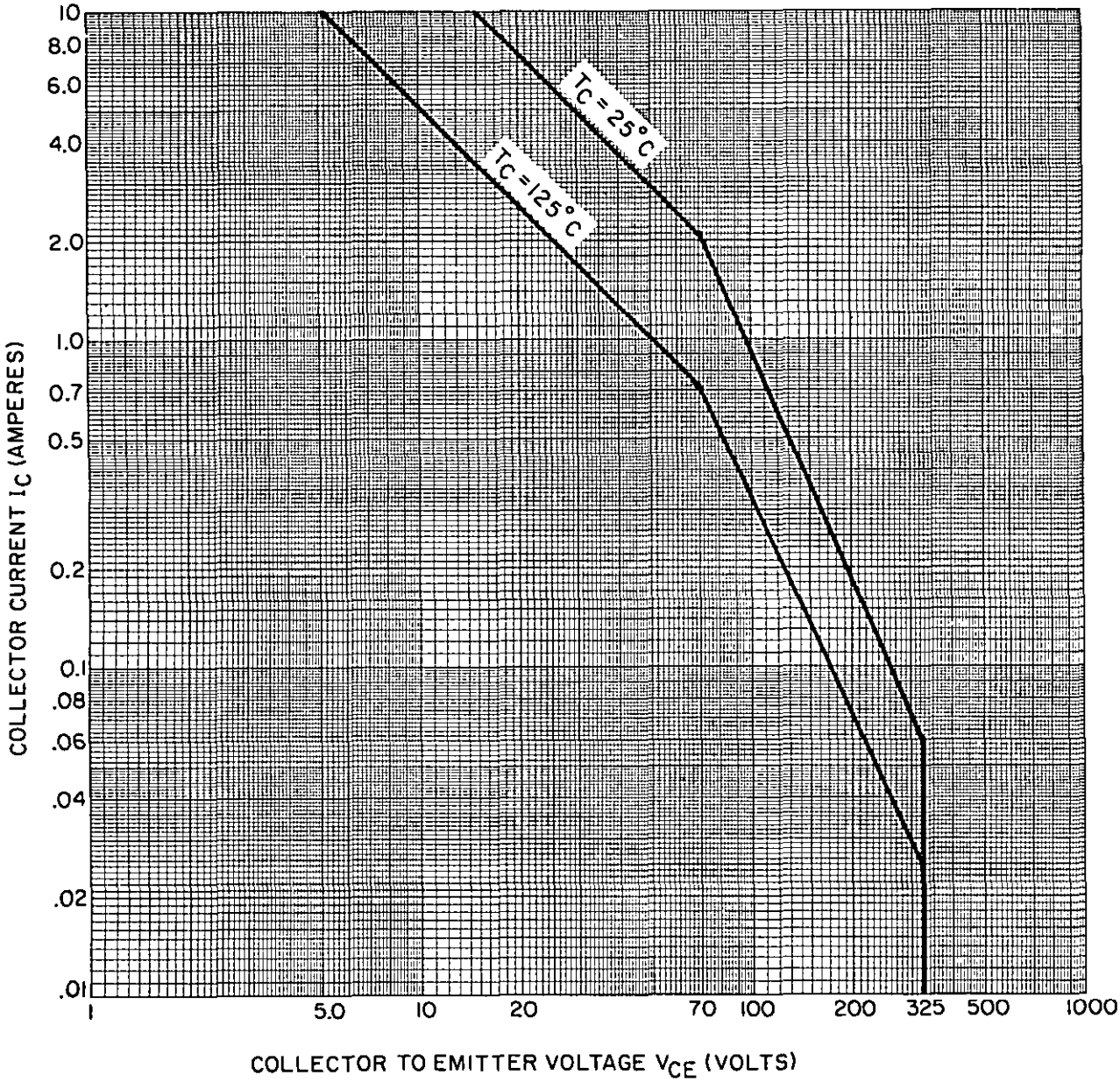


FIGURE 4. Maximum safe operating graph (continuous DC).

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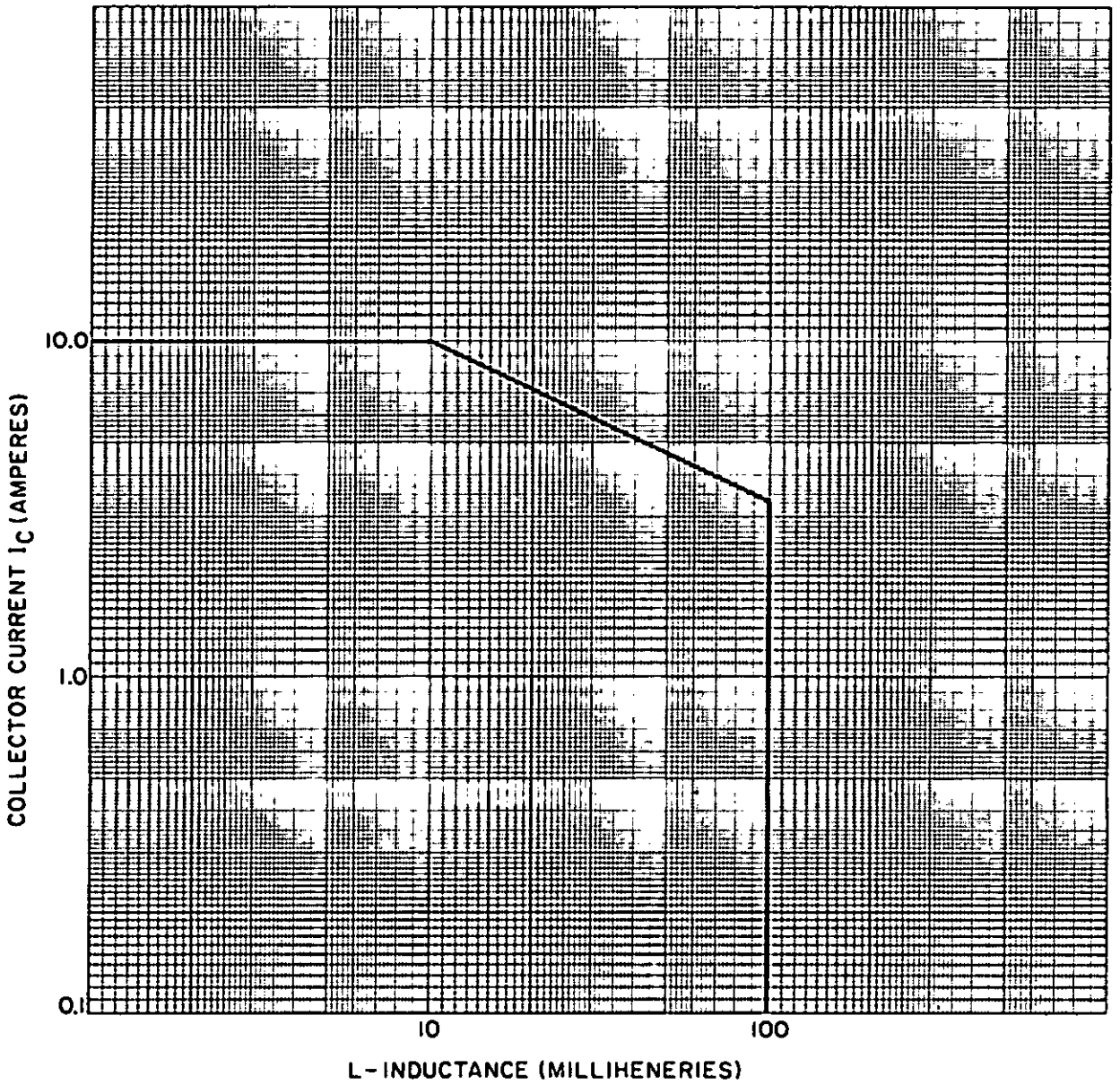


FIGURE 5. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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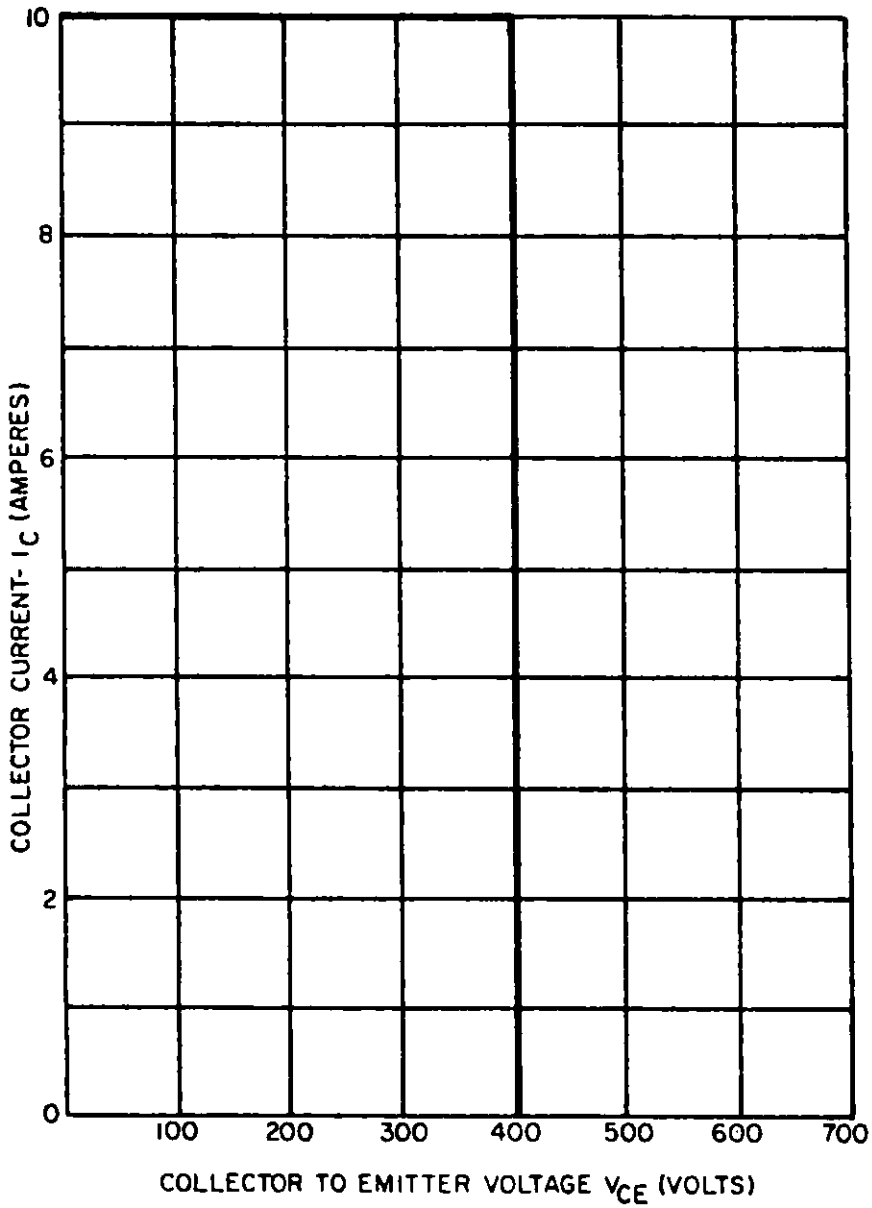
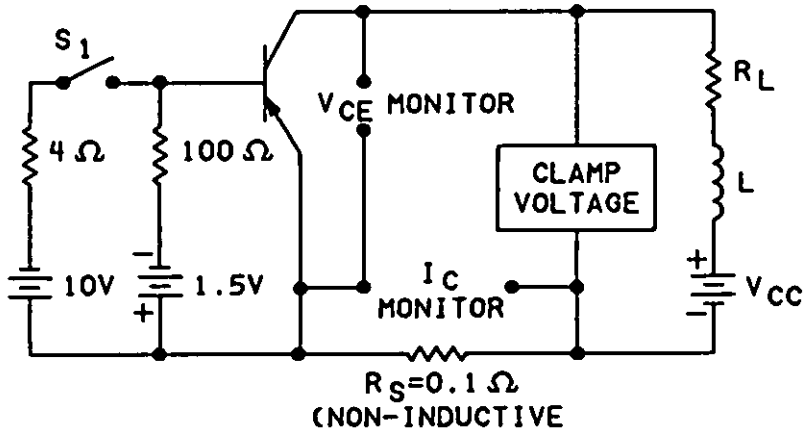


FIGURE 6. Safe operating area for switching between saturation and cutoff - (clamped inductive load).

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L = 10 mH, R ≈ 0.25Ω, Q ≥ 150 @ 1 KHz
f_r ≈ 100 KHz

Procedure:
1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached.
3. Perform specified end point tests.

FIGURE 7. Clamped inductive sweep test circuit.

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4.5.3 Group C life test. The group C sample (T_C test conditions) shall be drawn from the devices which were drawn for the group B sample (T_J test conditions).

4.5.4 Inspection conditions. Unless otherwise specified in MIL-S-19500 or herein, all inspections shall be conducted at a case temperature (T_C) of 25°C ±3°C.

4.5.5 Constant acceleration. Only devices constructed with aluminum bonding wire which successfully pass three consecutive lots with zero failures shall not require this test as a 100 percent screen on future inspection lots. In the event the sample fails the LTPD requirements of group B inspection, and the failures are catastrophic, (i.e., open or shorted condition), as verified by failure analysis, then 100 percent screening shall be re-instituted. The 100 percent screen shall be required on the failed lot and following lots until three successive inspection lots pass the constant acceleration test with zero failures. Only then may sampling inspection be resumed. End point measurements for this screening test shall be those required by screen 11. Only device failures, verified by failure analysis, caused by the constant acceleration test shall be counted as failures for this screening test. All sample devices which are drawn for constant acceleration inspection shall have successfully completed thermal shock inspection (minimum of 20 cycles).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Acquisition documents should specify the following:

a. Lead material and finish (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AV, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 15

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5961-0867)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19500/414B	Semiconductor Device, Transistor, NPN, Silicon High-Power Type 2N5241, JAN, JANTX, and JANTXV

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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